

Wasserstein Distances and Uniform Laws of Large Numbers

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1 Wasserstein Distances and Information Inequalities

We begin by completing the discussion of distances and divergences between probability measures. Let $(\mathcal{X}, \rho)$ be a metric space, and let $\mathbb{Q}$ and $\mathbb{P}$ be probability distributions on $\mathcal{X}$.

1.1 Optimal Transport Theory

Optimal transport theory studies how to transform one probability distribution $\mathbb{Q}$ into another probability distribution $\mathbb{P}$ while minimizing the cost of transporting probability mass.

A *transportation plan*, or *coupling*, of $\mathbb{Q}$ and $\mathbb{P}$ is a joint probability distribution M of a pair (X, X') such that

$$X \sim \mathbb{Q} \quad \text{and} \quad X' \sim \mathbb{P}.$$

The collection of all couplings of $\mathbb{Q}$ and $\mathbb{P}$ is denoted by $\Pi(\mathbb{Q}, \mathbb{P})$.

Let $\rho(x, x')$ denote the cost of moving one unit of mass from x to x' . For a coupling $M \in \Pi(\mathbb{Q}, \mathbb{P})$, its expected transportation cost is

$$\mathbb{E}_M [\rho(X, X')].$$

Definition 1.1 (Wasserstein distance). For $s \geq 1$, the order- s Wasserstein distance is

$$W_{\rho, s}(\mathbb{Q}, \mathbb{P}) := \inf_{M \in \Pi(\mathbb{Q}, \mathbb{P})} (\mathbb{E}_M [\rho(X, X')^s])^{1/s}. \quad (1)$$

When $s = 1$, we write $W_{\rho}(\mathbb{Q}, \mathbb{P}) := W_{\rho, 1}(\mathbb{Q}, \mathbb{P})$. The case $s = 2$ is also frequently used in practice.

Example 1.2 (Transporting bread to coffee stores). Suppose that bread is produced at two store locations, represented by -1 and 1 . The probability measure

$$\mathbb{Q} = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$$

describes the distribution of the bread supply. Thus, one half of the bread is produced at location -1 , and the other half is produced at location 1 .

The bread must be delivered to coffee stores at the same two locations. The probability measure

$$\mathbb{P} = \frac{1}{3}\delta_{-1} + \frac{2}{3}\delta_1$$

describes the distribution of demand. The coffee store at location -1 requires one third of the total bread, whereas the coffee store at location 1 requires two thirds.

Let $m(x, x')$ denote the amount of bread transported from production location x under $\mathbb{Q}$ to coffee-store location x' under $\mathbb{P}$. The four possible transportation quantities are

$$m(-1, -1), \quad m(-1, 1), \quad m(1, -1), \quad m(1, 1).$$

They have the following interpretations:

- $m(-1, -1)$ is the amount transported from the bread store at -1 to the coffee store at -1 ;
- $m(-1, 1)$ is the amount transported from the bread store at -1 to the coffee store at 1 ;
- $m(1, -1)$ is the amount transported from the bread store at 1 to the coffee store at -1 ;
- $m(1, 1)$ is the amount transported from the bread store at 1 to the coffee store at 1 .

The supply and demand constraints are

$$\begin{aligned} m(-1, -1) + m(-1, 1) &= \frac{1}{2}, & m(1, -1) + m(1, 1) &= \frac{1}{2}, \\ m(-1, -1) + m(1, -1) &= \frac{1}{3}, & m(-1, 1) + m(1, 1) &= \frac{2}{3}. \end{aligned}$$

An optimal transportation plan is

$$\begin{aligned} m(-1, -1) &= \frac{1}{3}, & m(-1, 1) &= \frac{1}{6}, \\ m(1, -1) &= 0, & m(1, 1) &= \frac{1}{2}. \end{aligned}$$

Equivalently, the plan is represented by the table

	Coffee store -1	Coffee store 1	Bread supply
Bread store -1	$m(-1, -1) = \frac{1}{3}$	$m(-1, 1) = \frac{1}{6}$	$\frac{1}{2}$
Bread store 1	$m(1, -1) = 0$	$m(1, 1) = \frac{1}{2}$	$\frac{1}{2}$
Coffee-store demand	$\frac{1}{3}$	$\frac{2}{3}$	1

Suppose that $\rho(x, x') = |x - x'|$. Transporting bread between stores at the same location has zero cost, whereas transporting bread from one location to the other has cost 2. Therefore, the total cost is

$$\begin{aligned} &\rho(-1, -1)m(-1, -1) + \rho(-1, 1)m(-1, 1) \\ &\quad + \rho(1, -1)m(1, -1) + \rho(1, 1)m(1, 1) \\ &= 0 \left(\frac{1}{3} \right) + 2 \left(\frac{1}{6} \right) + 2(0) + 0 \left(\frac{1}{2} \right) = \frac{1}{3}. \end{aligned}$$

The marginal constraints force at least $1/6$ of the bread produced at -1 to be transported to 1 . Hence no coupling can have a smaller cost, and

$$W_\rho(\mathbb{Q}, \mathbb{P}) = \frac{1}{3}.$$

For any 1-Lipschitz function f and any coupling $M \in \Pi(\mathbb{Q}, \mathbb{P})$,

$$\begin{aligned} \int_{\mathcal{X}} f d\mathbb{Q} - \int_{\mathcal{X}} f d\mathbb{P} &= \mathbb{E}_M [f(X) - f(X')] \\ &\leq \mathbb{E}_M [\rho(X, X')]. \end{aligned}$$

Taking the supremum over all 1-Lipschitz functions and then the infimum over all couplings gives the weak-duality inequality.

Theorem 1.3 (Kantorovich–Rubinstein duality). *Under the usual regularity and finite-first-moment assumptions,*

$$\begin{aligned} W_\rho(\mathbb{Q}, \mathbb{P}) &= \sup_{\|f\|_{\text{Lip}} \leq 1} \left[\int_{\mathcal{X}} f d\mathbb{Q} - \int_{\mathcal{X}} f d\mathbb{P} \right] \\ &= \inf_{M \in \Pi(\mathbb{Q}, \mathbb{P})} \int_{\mathcal{X} \times \mathcal{X}} \rho(x, x') dM(x, x') \\ &= \inf_{M \in \Pi(\mathbb{Q}, \mathbb{P})} \mathbb{E}_M [\rho(X, X')]. \end{aligned} \tag{2}$$

If $\mathbb{Q}$ and $\mathbb{P}$ have densities q and p , and the coupling M has density m , then the transportation cost associated with m is

$$\int_{\mathcal{X} \times \mathcal{X}} \rho(x, x') m(x, x') dx dx'.$$

Thus, the primal formulation (coupling formulation) minimizes transportation cost, whereas the dual formulation (Lipschitz function formulation) compares expectations over 1-Lipschitz functions.

1.2 Kullback–Leibler Divergence

Definition 1.4 (Kullback–Leibler divergence). The Kullback–Leibler (KL) divergence of $\mathbb{Q}$ from $\mathbb{P}$ is

$$D(\mathbb{Q} \parallel \mathbb{P}) := \begin{cases} \mathbb{E}_{\mathbb{Q}} \left[\log \frac{d\mathbb{Q}}{d\mathbb{P}} \right], & \mathbb{Q} \ll \mathbb{P}, \\ +\infty, & \text{otherwise.} \end{cases} \quad (3)$$

If $\mathbb{Q}$ and $\mathbb{P}$ have densities q and p with respect to a common dominating measure ν , then

$$D(\mathbb{Q} \parallel \mathbb{P}) = \int_{\mathcal{X}} q(x) \log \frac{q(x)}{p(x)} \nu(dx). \quad (4)$$

Unlike a metric, KL divergence is generally asymmetric and does not satisfy the triangle inequality. It nevertheless controls total variation distance.

Theorem 1.5 (Pinsker–Csiszár–Kullback inequality). *With the convention*

$$\|\mathbb{P} - \mathbb{Q}\|_{\text{TV}} := \sup_{A \subseteq \mathcal{X}} |\mathbb{P}(A) - \mathbb{Q}(A)|,$$

we have

$$\|\mathbb{P} - \mathbb{Q}\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D(\mathbb{Q} \parallel \mathbb{P})}. \quad (5)$$

Thus, small KL divergence implies that the two measures assign similar probabilities to every measurable event.

2 Uniform Laws of Large Numbers

2.1 Uniform Convergence of CDFs

Let X be a real-valued random variable with cumulative distribution function

$$F(t) := \mathbb{P}(X \leq t) = \mathbb{E} [\mathbf{1}_{\{X \leq t\}}], \quad t \in \mathbb{R}.$$

Suppose that

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Law}(X).$$

Definition 2.1 (Empirical CDF). The empirical cumulative distribution function is

$$\hat{F}_n(t) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq t\}}, \quad t \in \mathbb{R}. \quad (6)$$

For every fixed t , the strong law of large numbers gives

$$\widehat{F}_n(t) \longrightarrow F(t) \quad \text{almost surely.}$$

This is pointwise convergence: the exceptional event may initially depend on t .

Many estimation problems can be formulated through a functional γ that maps a CDF F to a real number $\gamma(F)$. The plug-in estimator is $\gamma(\widehat{F}_n)$, and the basic consistency question is whether

$$\gamma(\widehat{F}_n) \xrightarrow{?} \gamma(F).$$

Example 2.2 (Expectation functionals). Let g be an integrable function and define

$$\gamma_g(F) := \int_{\mathbb{R}} g(x) dF(x).$$

Because $\widehat{F}_n$ assigns mass $1/n$ to each observation,

$$\gamma_g(\widehat{F}_n) = \int_{\mathbb{R}} g(x) d\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n g(X_i).$$

Therefore, if $\mathbb{E}|g(X)| < \infty$, the strong law gives

$$\gamma_g(\widehat{F}_n) \longrightarrow \gamma_g(F) \quad \text{almost surely.}$$

Example 2.3 (Quantile functionals). For $\alpha \in (0, 1)$, the lower α -quantile functional is

$$Q_\alpha(F) := \inf \{t \in \mathbb{R} : F(t) \geq \alpha\}.$$

The corresponding empirical quantile is

$$\begin{aligned} Q_\alpha(\widehat{F}_n) &= \inf \left\{ t \in \mathbb{R} : \widehat{F}_n(t) \geq \alpha \right\} \\ &= \inf \left\{ t \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq t\}} \geq \alpha \right\}. \end{aligned}$$

It is not immediately clear from pointwise convergence alone whether $Q_\alpha(\widehat{F}_n) \rightarrow Q_\alpha(F)$. Uniform convergence and an appropriate condition ensuring that F crosses the level α are needed.

Definition 2.4 (Supremum norm and continuity). For two CDFs F and G , their supremum distance is

$$\|F - G\|_\infty := \sup_{t \in \mathbb{R}} |F(t) - G(t)|. \quad (7)$$

A functional γ is continuous at F in the supremum norm if, for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\|G - F\|_\infty \leq \delta \implies |\gamma(G) - \gamma(F)| \leq \epsilon.$$

Recall that if f is a continuous function on $\mathbb{R}^d$, then

$$x_n \rightarrow x \implies f(x_n) \rightarrow f(x).$$

Similarly, if $F_n \rightarrow F$ in the supremum norm and γ is continuous at F , then

$$\gamma(F_n) \rightarrow \gamma(F).$$

Therefore, to establish the consistency of $\gamma(\widehat{F}_n)$, it is sufficient to verify that

1. $\|\hat{F}_n - F\|_\infty \rightarrow 0$;
2. γ is continuous at F with respect to the supremum norm.

Theorem 2.5 (Glivenko–Cantelli). *For every distribution function F , the empirical CDF $\hat{F}_n$ is a strongly consistent estimator of the population CDF in the supremum norm:*

$$\|\hat{F}_n - F\|_\infty \longrightarrow 0 \quad \text{almost surely.} \quad (8)$$

The proof is deferred.

2.2 Uniform Laws for General Function Classes

The empirical-CDF result is a special case of a general problem involving empirical averages over a collection of functions. Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}$$

on a measurable space $\mathcal{X}$.

Definition 2.6 (Empirical measure). The empirical probability measure is

$$\mathbb{P}_n := \frac{1}{n} \sum_{i=1}^n \delta_{X_i}. \quad (9)$$

For $a \in \mathcal{X}$, the Dirac probability measure δ_a is defined by

$$\delta_a(B) := \begin{cases} 1, & a \in B, \\ 0, & a \notin B, \end{cases} \quad B \subseteq \mathcal{X}. \quad (10)$$

Example 2.7 (Bernoulli distribution). The Bernoulli distribution with parameter p can be written as

$$\text{Ber}(p) = p\delta_1 + (1-p)\delta_0.$$

Its probability mass function is

$$\text{pmf}(x) = \begin{cases} p, & x = 1, \\ 1-p, & x = 0. \end{cases}$$

Thus, if $\mathbb{P} = p\delta_1 + (1-p)\delta_0$, then

$$\mathbb{P}(\{1\}) = p, \quad \mathbb{P}(\{0\}) = 1-p.$$

Definition 2.8 (Population and empirical averages). For a measurable function f , define

$$\mathbb{P}f := \int_{\mathcal{X}} f(x) d\mathbb{P}(x) = \mathbb{E}_{\mathbb{P}}[f(X)]. \quad (11)$$

The corresponding empirical average is

$$\mathbb{P}_n f := \int_{\mathcal{X}} f(x) d\mathbb{P}_n(x) = \frac{1}{n} \sum_{i=1}^n f(X_i). \quad (12)$$

For each fixed integrable function f , the strong law of large numbers implies

$$\mathbb{P}_n f \longrightarrow \mathbb{P}f \quad \text{almost surely.}$$

Definition 2.9. Let $\mathcal{F}$ be a class of measurable functions. Define

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} := \sup_{f \in \mathcal{F}} |\mathbb{P}_n f - \mathbb{P} f|. \quad (13)$$

The class $\mathcal{F}$ is called a $\mathbb{P}$ -Glivenko–Cantelli class if

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \longrightarrow 0 \quad \text{almost surely.} \quad (14)$$

Example 2.10 (Empirical CDFs and indicator functions). Define the threshold class

$$\mathcal{F}_{\text{cdf}} := \{f_t(\cdot) : t \in \mathbb{R}\}, \quad f_t(x) := \mathbf{1}_{\{x \leq t\}}.$$

For every $t \in \mathbb{R}$,

$$\mathbb{P} f_t = \int_{\mathcal{X}} \mathbf{1}_{\{x \leq t\}} d\mathbb{P}(x) = \mathbb{P}(X \leq t) = F(t),$$

and

$$\mathbb{P}_n f_t = \frac{1}{n} \sum_{i=1}^n f_t(X_i) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq t\}} = \widehat{F}_n(t).$$

Therefore,

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}_{\text{cdf}}} = \sup_{t \in \mathbb{R}} \left| \widehat{F}_n(t) - F(t) \right| = \left\| \widehat{F}_n - F \right\|_{\infty}.$$

The Glivenko–Cantelli theorem says precisely that the threshold class $\mathcal{F}_{\text{cdf}}$ is a universal Glivenko–Cantelli class.

This reformulation is the bridge from empirical CDFs to empirical-process theory. For a single function, the law of large numbers is enough. For an entire class, the size and structure of the class determine whether the same sample controls all empirical averages simultaneously.